

# Termo IB - Resumen de fórmulas

## 1. Gases ideales

$$PV = nRT \quad PM = \rho RT, \quad \rho \equiv \text{densidad}, \mathcal{M} \equiv \text{Masa molar} \quad (1)$$

## 2. Conservación de la energía

$$dE = dU + \sum dE_i = \delta Q - \delta W$$

$$dU = n c_v dT \quad dH = dU + d(PV)$$

Si  $P = \text{cte}$ ,  $dH = dU + PdV = n c_p dT$

## 3. Transformaciones adiabáticas cuasiestáticas en gases

$$c_v = \{3/2R, 5/2R, 3R\} \quad c_p = c_v + R$$

$$\gamma = \frac{c_p}{c_v} \quad T V^{\gamma-1} = c_1; T P^{(1-\gamma)/\gamma} = c_2; P V^\gamma = c_3;$$

$$c_p = \frac{\gamma}{\gamma-1} R \quad c_v = \frac{1}{\gamma-1} R$$

## 4. Sistemas abiertos

$$h^0 = h + gz + \frac{1}{2}c^2$$

$$dE_{VC} = \delta Q + \sum_k h_{in}^0 |\delta m| - \sum_k h_{out}^0 |\delta m| - \sum_k \delta W_k - \delta W_{ExpVC}$$

$$\delta m_{VC} = \sum_k |\delta m_{in}| - \sum_k |\delta m_{out}|$$

Bernoulli:  $P_2 + \rho g z_2 + \frac{1}{2} \rho c_2^2 = P_1 + \rho g z_1 + \frac{1}{2} \rho c_1^2 = \text{cte}$

## 5. Dispositivos en régimen estacionario

Bomba (líquido) -  $w_{eje} = h_2 - h_1 = \Delta u + v \Delta P$

Turbina:  $w_{eje} = h_{in} - h_{out} = c_p (T_{in} - T_{out})$

Intercambiador de calor:  $\sum_k h_{out}^0 \dot{m}_{out} = \sum_k h_{in}^0 \dot{m}_{in}$

Tobera-Difusor:

$$h_1^+ = h_1 + \frac{1}{2} c_1^2 = h_2^+ \quad \dot{m} = c_1 A_1 \rho_1 = c_2 A_2 \rho_2 = \text{cte} \quad (2)$$

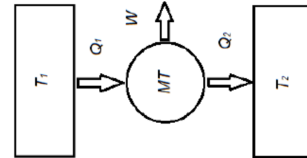
## 6. Segundo principio

Máquina Térmica:  $W = Q_1 - Q_2$

$$\eta_{MT} = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} \Rightarrow 1 - \frac{T_2}{T_1}$$

Máquina Frigorífica:  $Q_1 = Q_2 + W \quad \beta_{MF} = \frac{Q_2}{W}$

$$\beta_{BC} = \frac{Q_1}{W} = \frac{Q_2 + W}{W}$$



Entropía:  $dS = \frac{\delta Q_{rev}}{T} \quad dS_{Real} \geq 0 \quad dS = \frac{\delta Q}{T} + dS_{gen}$

$$ds = \frac{du}{T} + \frac{Pdv}{T} = \frac{dh}{T} - \frac{v dP}{T}$$

Líquidos/Sólidos incompresibles:  $c_v = c_p = C, dv = 0$

$$ds = \frac{CdT}{T}$$

Gases ideales:  $ds = \frac{c_v dT}{T} + \frac{Rdv}{v} = \frac{c_p dT}{T} - \frac{RdP}{P}$

### 6.1. Sistemas abiertos con j entradas y salidas y k fuentes de temperatura constante:

$$dS_{VC} \geq \sum_j \delta m_j + \sum_k \frac{\delta Q_k}{T_k}$$

Escrito por unidad de tiempo en régimen estacionario:

$$-\sum_j s_j \dot{m}_j = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen}$$

Trabajo para una máquina en flujo estable:

$$w_{rev} = - \int_1^2 v dP - \Delta k_E - \Delta p_E$$

### 6.2. Eficiencias isoentrópicas

$$\eta_{Bomba}^S = \frac{W_{Rev}}{W_{Real}} = \frac{v \Delta P}{c \Delta T + v \Delta P}$$

$$\eta_{Compresor}^S = \frac{W_{Rev}}{W_{Real}} = \frac{h_{out}^s - h_{in}}{h_{out} - h_{in}}$$

$$\text{Gases Ideales Perf.: } \eta_{Compresor}^S = \frac{T_{out}^s - T_{in}}{T_{out} - T_{in}}$$

Donde  $T_{out}^s$  se obtiene mediante las relaciones de Poisson.

$$\eta_{Tobera}^S = \left( \frac{c_{out}}{c_{out}^s} \right)^2 = \frac{h_{out}^+ - h_{out}}{h_{out}^+ - h_{out}^s}$$

## 7. Exergía

Exergía:  $X = m\Phi = E - E_0 - T_0(S - S_0) + P_0(V - V_0) = -W_{ú,máx}$

$$W_{útil} = W - W_{ATM} = W - P_0(V - V_0)$$

$$\text{Flow Exergy: } \Psi = (h - h_0) - T_0(s - s_0) + \frac{s^2}{2} + gz$$

Sistema A que intercambia calor con k fuentes a  $T_K$ :

$$\Delta\Phi = \Delta EX = \Delta E_A - T_0\Delta S_A + P_0\Delta V_A + \sum_k Q_k \left(1 - \frac{T_0}{T_k}\right)$$

$$\text{Irreversibilidad: } I = T_0 S_{gen} = T_0 \left( \Delta S_A - \sum_k \frac{Q_k}{T_k} \right)$$

Sistema dinámico:

$$\sum \left(1 - \frac{T_0}{T_k}\right) \delta Q_k - [W - P_0 dV] +$$

$$\sum \delta m_i \Psi_i - \delta m_e \Psi_e - X_{destr} = dX$$

## 8. Potencial químico

$$dU = TdS - PdV + \sum_j \mu_j dn_j \quad \mu_j = \left. \frac{\partial U}{\partial n_j} \right|_{S,V,n_k \neq n_j}$$

$$dH = TdS + VdP + \sum_j \mu_j dn_j \quad \mu_j = \left. \frac{\partial H}{\partial n_j} \right|_{S,P,n_k \neq n_j}$$

Energía libre de Helmholtz:  $F = U - TS$

$$dF = -SdT + PdV + \sum_j \mu_j dn_j \quad \mu_j = \left. \frac{\partial F}{\partial n_j} \right|_{T,V,n_k \neq n_j}$$

Energía libre de Gibbs:  $H - TS$

$$dG = -SdT + VdP + \sum_j \mu_j dn_j \quad \mu_j = \left. \frac{\partial G}{\partial n_j} \right|_{T,P,n_k \neq n_j}$$

Ecuación de Gibbs-Duhem:

$$SdT - VdP + \sum_j n_j d\mu_j = 0$$

$$\mu = g = h - Ts, \quad h - h_0 = \int_{T_0}^T c_P(T) dT$$

$$s - s_0 = \int_T^{T_0} \frac{c_P(T)}{T} dT - R \int_P^{P_0} \frac{dP}{P}$$

$$\mu = \mu(T, P) = \varphi(T) + RT \ln \left( \frac{P}{P_0} \right)$$

Condición de equilibrio de mezcla:  $d\mu_j = d\mu_k$

$$\text{Ley de Dalton: } P_j = \frac{n_j}{V} RT = y_j P_T, \quad y_j = \frac{n_j}{n_T} \quad ()_T = [N]_{\text{Total}}$$

## 8.1. Entropía de mezcla

$$\Delta S = n \sum_j y_j \Delta s_j$$

$$\text{Gases ideales: } \Delta s_j = c_{Pj} \ln \left( \frac{T_2}{T_1} \right) - R \ln \left( \frac{y_{j,2} P_2}{y_{j,1} P_1} \right),$$

Donde  $T_k$  y  $P_k$  son las temperaturas y presiones de cada estado y cantidad de gas ideal.

## 9. Lagrangiano del modelo estándar

$$\begin{aligned}
\mathcal{L} = & -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}\text{tr}(\mathbf{W}_{\mu\nu}\mathbf{W}^{\mu\nu}) - \frac{1}{4}\text{tr}(\mathbf{G}_{\mu\nu}\mathbf{G}^{\mu\nu}) && \text{(gauge terms: U(1), SU(2), SU(3))} \\
& + (\bar{\nu}_L, \bar{e}_L) \tilde{\sigma}^\mu i D_\mu \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} + \bar{e}_R \sigma^\mu i D_\mu e_R + \bar{\nu}_R \sigma^\mu i D_\mu \nu_R + (\text{h.c.}) && \text{(lepton kinetic terms)} \\
& - \frac{\sqrt{2}}{\nu} \left[ (\bar{\nu}_L, \bar{e}_L) \phi M^e e_R + \bar{e}_R M^{e\dagger} \phi^\dagger \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} \right] && \text{(charged lepton mass term)} \\
& - \frac{\sqrt{2}}{\nu} \left[ (-\bar{e}_L, \bar{\nu}_L) \phi^* M^\nu \nu_R + \bar{\nu}_R M^{\nu\dagger} \phi^T \begin{pmatrix} -e_L \\ \nu_L \end{pmatrix} \right] && \text{(neutrino mass term)} \\
& + (\bar{u}_L, \bar{d}_L) \tilde{\sigma}^\mu i D_\mu \begin{pmatrix} u_L \\ d_L \end{pmatrix} + \bar{u}_R \sigma^\mu i D_\mu u_R + \bar{d}_R \sigma^\mu i D_\mu d_R + (\text{h.c.}) && \text{(quark kinetic terms)} \\
& - \frac{\sqrt{2}}{\nu} \left[ (\bar{u}_L, \bar{d}_L) \phi M^d d_R + \bar{d}_R M^{d\dagger} \phi^\dagger \begin{pmatrix} u_L \\ d_L \end{pmatrix} \right] && \text{(down-type quark mass term)} \\
& - \frac{\sqrt{2}}{\nu} \left[ (-\bar{d}_L, \bar{u}_L) \phi^* M^u u_R + \bar{u}_R M^{u\dagger} \phi^T \begin{pmatrix} -d_L \\ u_L \end{pmatrix} \right] && \text{(up-type quark mass term)} \\
& + (D_\mu \phi)^\dagger D^\mu \phi - \frac{m_h^2}{2\nu^2} \left( \phi^\dagger \phi - \frac{\nu^2}{2} \right)^2 && \text{(Higgs kinetic + potential)}
\end{aligned}$$